

Functional & Performance Testing Template

Model Performance Test

Date	18 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	

Test Scenarios & Results — AI Nutrition Chatbot

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Nutrition question response	Send a valid nutrition question (e.g. "How do I lose weight healthily?") to POST /api/ai/chat	A relevant, markdown-formatted response is returned, either generated by Gemini or the keyword-based fallback		
FT-02	Empty message validation	Submit a chat request with an empty message field	API responds with 400 and "Message is required"		
FT-03	Fallback behaviour without API key	Omit GEMINI_API_KEY from the server environment and send a message	A smart mock response matching the message keywords (weight loss, recipe, water, diet) is returned instead of an error		
FT-04	Conversation history handling	Send a message along with a prior history array of previous turns	The reply reflects the conversational context passed in history		
PT-01	Response Time Test	Time how long it takes to receive a chatbot reply after submitting a message	Should be under 3–5 seconds under normal network conditions		
PT-02	Concurrent Chat Requests Test	Submit multiple chat requests to /api/ai/chat at the same time	The service should not slow down significantly or fail under concurrent load		
PT-03	Long Conversation History Load Test	Send a request with a long history array	Should process without crashing, respecting the maxOutputTokens limit configured on the Gemini request		